

修士論文概要書

Master's Thesis Summary

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研究指導名 Research guidance	暗号プロトコル 研究	学籍番号 Student ID number	5124FG15			
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1 Introduction

Post-quantum anonymous credentials let a holder prove possession of a valid credential without revealing identity, under assumptions that survive a quantum adversary. We target the setting in which the holder generates this proof on a *personal device*, here a consumer laptop (18-core, 24 GB), as opposed to a cloud or server prover. A general-purpose zero-knowledge virtual machine (zkVM) proves an arbitrary verification predicate without a bespoke, per-scheme circuit, at the cost of a generality tax on the arithmetic. The obstacle is arithmetic. The schemes we care about (PLUM [1], Loquat [2]) operate over a nominally 192-bit (concretely 199-bit) prime field, while production zkVMs (RISC Zero [3], SP1 [4]) fix a small (≈ 31 -bit) native field, so every operation is emulated by multi-limb arithmetic; an algebraic hash (Griffin [5]), which makes up $\approx 91\%$ of the verification's R1CS constraints at PLUM-128 [1, §4.2], amplifies the penalty. This thesis asks *when* a custom precompile pays for compiling such a signature into a zkVM; using that criterion it reduces the proving cost of the verification on a personal device, and it locates where a jointly zero-knowledge and post-quantum proof is still out of reach.

2 The Field-Mismatch Tax and When a Precompile Pays

We formalize the *field-mismatch tax*: the cost that arises when a primitive's natural algebraic domain (here a large prime field $\mathbb{F}_q$) does not coincide with the proof system's *native* arithmetic. A backend is modelled by its native field $\mathbb{F}_p$, an encoding of the computation into $\mathbb{F}_p$, and a constraint representation (R1CS for static circuits, AIR/trace for STARK-based zkVMs). In the *field-aligned* regime of application-specific circuits the designer matches $\mathbb{F}_p$ to the primitive's domain, so arithmetic is expressed directly and constraint growth tracks multiplicative complexity. In the *mismatch* regime of general-purpose zkVMs the native field is fixed and much smaller, so each $\mathbb{F}_q$ element is carried as $k = \lceil \log_2 q/w \rceil$ w -bit limbs with carry propagation and modular reduction.

A counting argument gives a lower bound: a single non-native multiplication forces $\Omega(k)$ native multiplications, since the product depends on all k input limbs while each native multiply carries at most one. For STARK provers this amplifies to a $\Theta(k^2)$ inflation of the schoolbook-emulated trace, and since

prover time scales roughly as $O(W T \log T)$ in trace width W and length T , the tax is a structural slowdown, not a constant-factor one. This yields the criterion: a precompile lowers cost only on the field-mismatch axis, and pays only when it removes more emulated trace area than the dedicated chip adds. A primitive already served by an existing precompile, or native to the prover field, therefore does not warrant a bespoke chip even though one could be built.

3 System Design

A *precompile* is a dedicated sub-circuit (an *AIR*: algebraic intermediate representation, the STARK trace's constraint form) for a designated primitive that the zkVM guest invokes as a syscall, so the primitive's cost bypasses field emulation. We construct one around a reusable interface and instantiate it as a small *family* of three chips over PLUM's 199-bit prime field: (i) a bespoke Griffin permutation [5] AIR, the load-bearing 91% target; (ii) a dedicated FP192_MUL field-multiplication chip; and (iii) a FP192_POW_RES chip composing that multiplication into the t -th power-residue PRF symbol check $a^{(p-1)/t} \bmod p$ ($t = 256$ fixed). All three build and pass a core-STARK smoke proof (execute, prove, verify), each with a written soundness argument tying its constraints to the reference specification, so the precompile layer preserves the scheme's unforgeability under the assumption that the zkVM's cross-table lookup argument is sound. The criterion then separates the three: the Griffin chip pays against emulation and recovers feasibility (§5), yet loses to a field-matched control; the field-multiplication chip does not pay, since the host's 256-bit modular-multiplication precompile already serves its arithmetic; and the power-residue symbol, the one irreducibly-large-prime primitive, pays $10.33\times$ on total committed trace area.

4 Evaluation

We measure four cells at $\lambda = 80$ (below the 128-bit deployment level, the largest that fits the 24 GB budget; Table 1). Cells 1–3 share one scheme, so with/without/control is a matched comparison; Cell 4 (Aurora [6]) is a different scheme over a different field, an indicative cross-scheme reference whose crossover is bounded in form, not located in wall-clock.

Table 1. Four cells (PLUM, $\lambda=80$, 24 GB, $n=5$ for cells 2–3).

Cell	Setting	Prove
1	no precompile	> 5 h
2	Griffin precompile	14.24 min
3	SHA-3 control	13.24 min
4	Aurora (Loquat)	≈22 min

With the precompile, verification fits the 24 GB budget for the first time in the study (peak ≈ 14 GB; Cell 1 does not terminate). The control is the sharpest result: even accelerated, PLUM verification proves $1.08\times$ slower and runs $1.13\times$ more cycles than the software SHA-3 (Table 1), so a standardized hash, unaccelerated, still beats our algebraic hash. The trace-area criterion postdicts the sign of this inversion, and a matched-field key-stone ($164\times$ at $\ell = 1$ versus $\ell = 7$ limbs, a field-aligned operand versus the emulated 199-bit one) fixes its necessity direction, so the inversion is evidence for the criterion, not a paradox.

5 Discussion

Feasibility is not deployability, and the gap is a *dual obstruction*. The feasible, post-quantum-sound proof, the bare STARK receipt, is not zero-knowledge (Horn 1); and the zero-knowledge wrap the toolchain ships, though it now fits the 24 GB budget, is a Groth16/PLONK SNARK over a pairing-friendly curve and hence not post-quantum (Horn 2), retroactively de-anonymizing every stored transcript to a future quantum adversary. No proof the stack produces is at once zero-knowledge and post-quantum-sound: the obstruction is a primitive limit, not feasibility. For the anonymous-credential construction BDEC [7] that PLUM instantiates, we further prove that *everlasting* anonymity cannot be established in the standard model for its hash-committed showings [8], escapable only by a post-quantum statistically-hiding commitment to the witness. The positive results are conditional, resting on five named open assumptions (Griffin-AIR soundness, transcript simulatability, the quantum-random-oracle lift, lookup binding, and signature key-privacy).

6 Conclusion

We formalized the field-mismatch tax and derived a criterion for when a custom precompile pays, built a family of three precompiles instantiating one interface (all three running) to test it, brought post-quantum anonymous-credential verification within the personal-device memory budget on this substrate, and located a *dual obstruction* that blocks a jointly post-quantum and zero-knowledge proof. Closing it, by masking the prover’s low-degree test with a post-quantum-sound wrap [9, 10] in place of the pairing-based one and pushing below the personal-device budget toward mobile, is left as future work.

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